Visualisation

July 31, 2023

APPLIED LINGUISTICS GRADUATE PROGRAMME (LAEL)

1 Visualisation Practice Exercise 1

1.1 Python Data Visualization with Seaborn and Matplotlib

This practice exercise was guided by this tutorial. The dataset is related to FIFA 19 players and can be obtained here.

1.1.1 Histogram

[1]:
```
# Importing the required libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
import seaborn as sns
```

[2]:
```
# Casting the FIFA 19 data in to the dataframe 'df'
# Please make sure the 'fifa_eda.csv'is in the folder
df = pd.read_csv('fifa_eda.csv')
print(df.dtypes)
df.head()
```

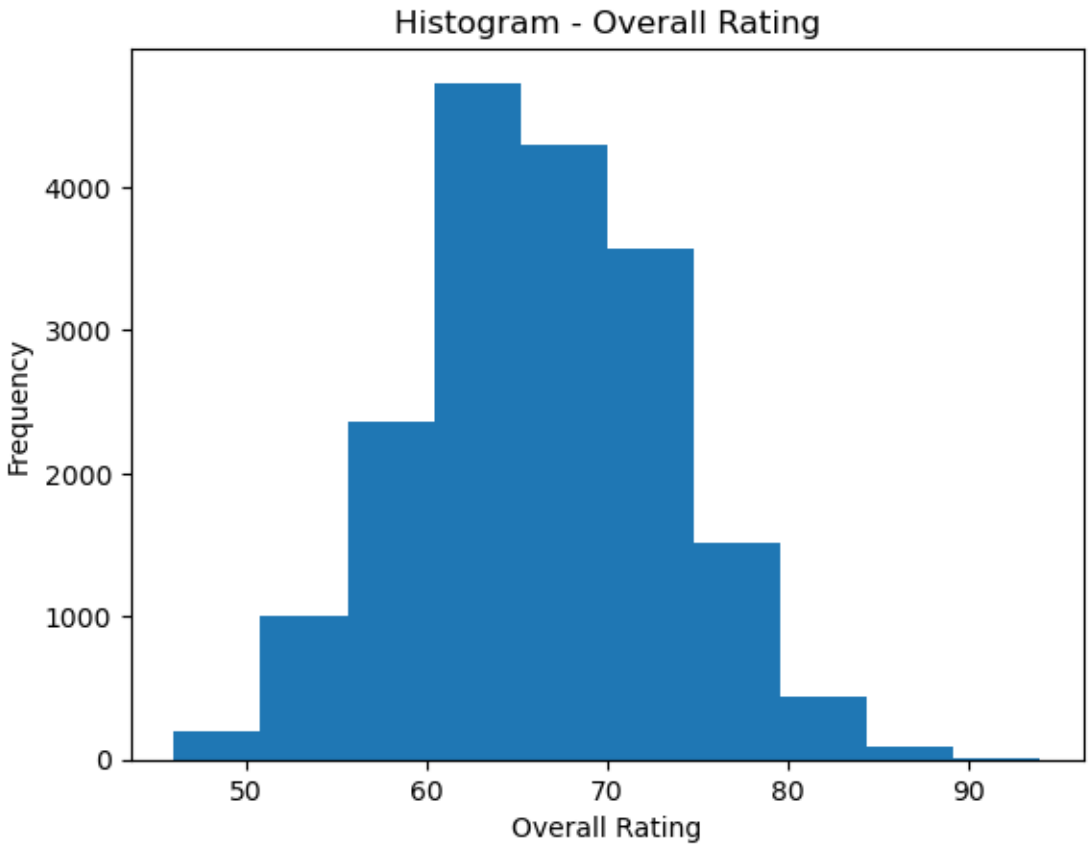
```
ID                          int64
Name                       object
Age                         int64
Nationality                object
Overall                     int64
Potential                   int64
Club                       object
Value                     float64
Wage                      float64
Preferred Foot             object
International Reputation  float64
Skill Moves               float64
Position                   object
Joined                      int64
```

```
Contract Valid Until         object
Height                      float64
Weight                      float64
Release Clause              float64
dtype: object
```

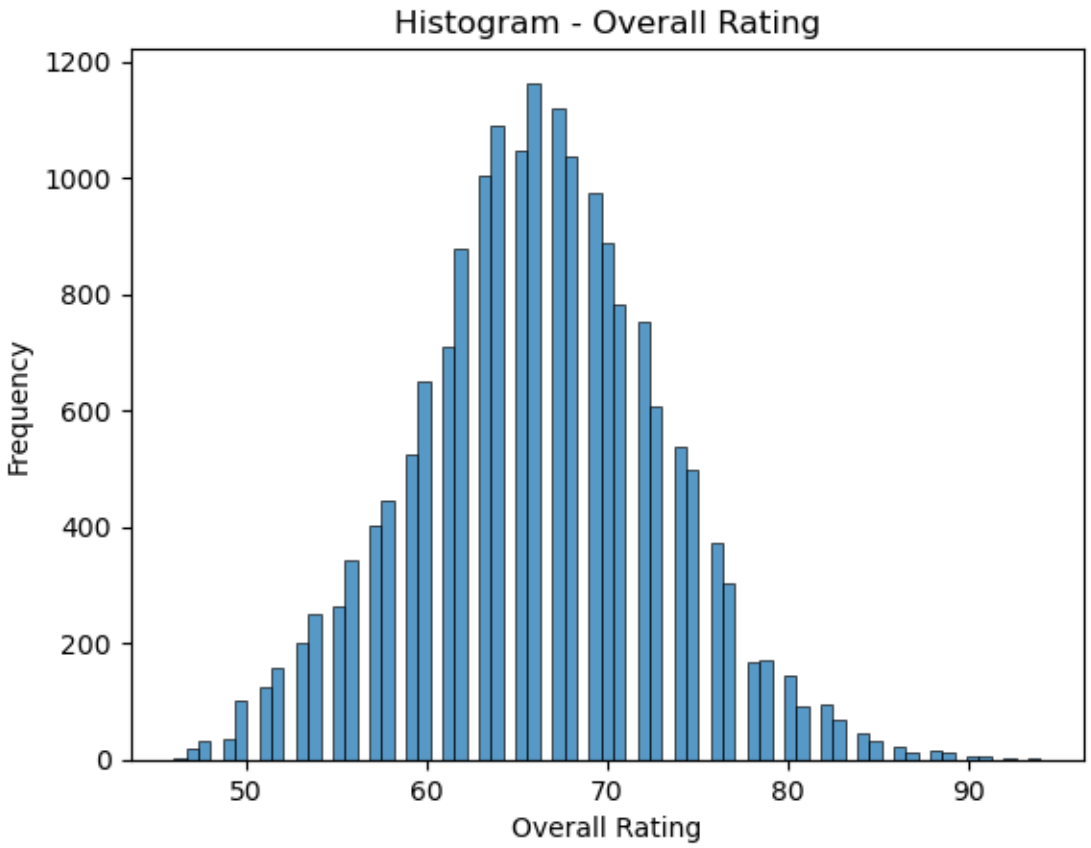
[2]:

	ID	Name	Age	Nationality	Overall	Potential	Club	Value	Wage	Preferred Foot	International Reputation	Skill Moves	Position	Joined	Contract Valid Until	Height	Weight	Release Clause
0	158023	L. Messi	31	Argentina	94	94	FC Barcelona	110500.0	565.0	Left	5.0	4.0	RF	2004	2021-01-01	5.583333	159.0	226500.0
1	20801	Cristiano Ronaldo	33	Portugal	94	94	Juventus	77000.0	405.0	Right	5.0	5.0	ST	2018	2022-01-01	6.166667	183.0	127100.0
2	190871	Neymar Jr	26	Brazil	92	93	Paris Saint-Germain	118500.0	290.0	Right	5.0	5.0	LW	2017	2022-01-01	5.750000	150.0	228100.0
3	193080	De Gea	27	Spain	91	93	Manchester United	72000.0	260.0	Right	4.0	1.0	GK	2011	2020-01-01	6.333333	168.0	138600.0
4	192985	K. De Bruyne	27	Belgium	91	92	Manchester City	102000.0	355.0	Right	4.0	4.0	RCM	2015	2023-01-01	5.916667	154.0	196400.0

[3]:
```
# Histogram of the 'Overall Rating' using Matplotlib
plt.hist(df[['Overall']])
plt.title('Histogram - Overall Rating')
plt.xlabel('Overall Rating')
plt.ylabel('Frequency')
plt.show()
```



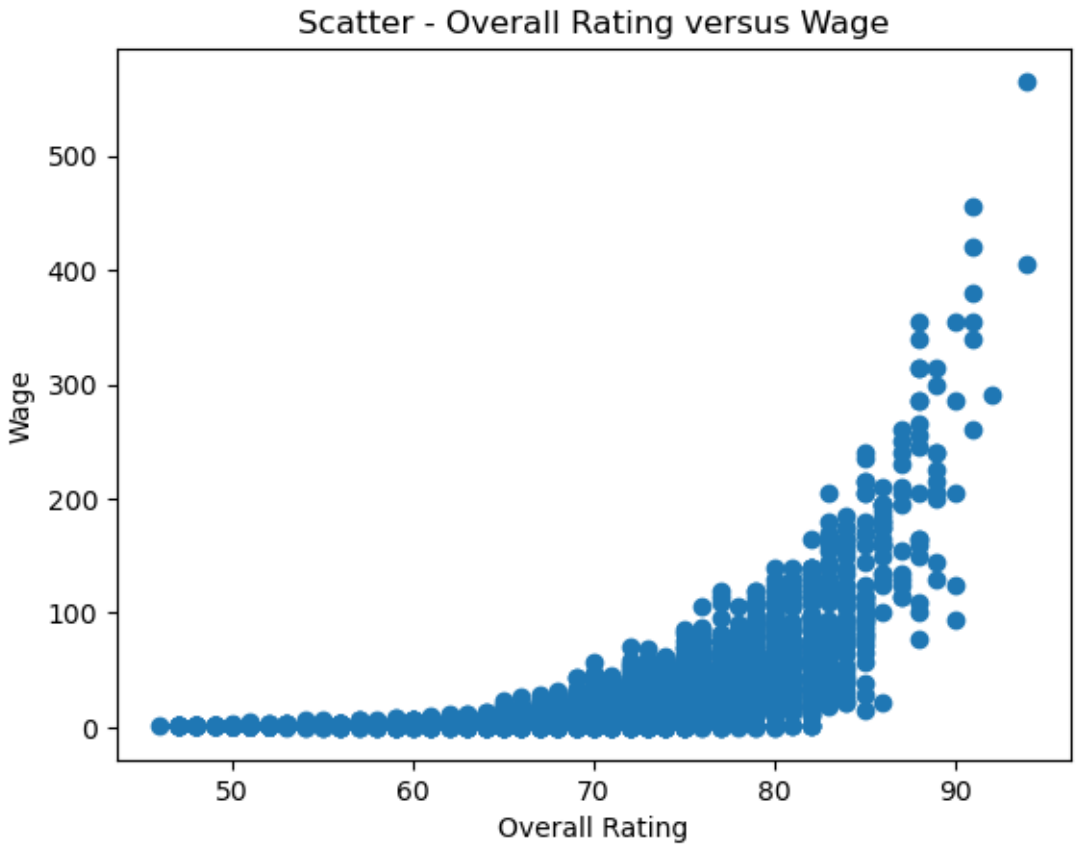
[4]:
```
# Histogram of the 'Overall Rating' using Seaborn
sns.histplot(data = df, x = 'Overall')
plt.title('Histogram - Overall Rating')
plt.xlabel('Overall Rating')
plt.ylabel('Frequency')
plt.show()
```



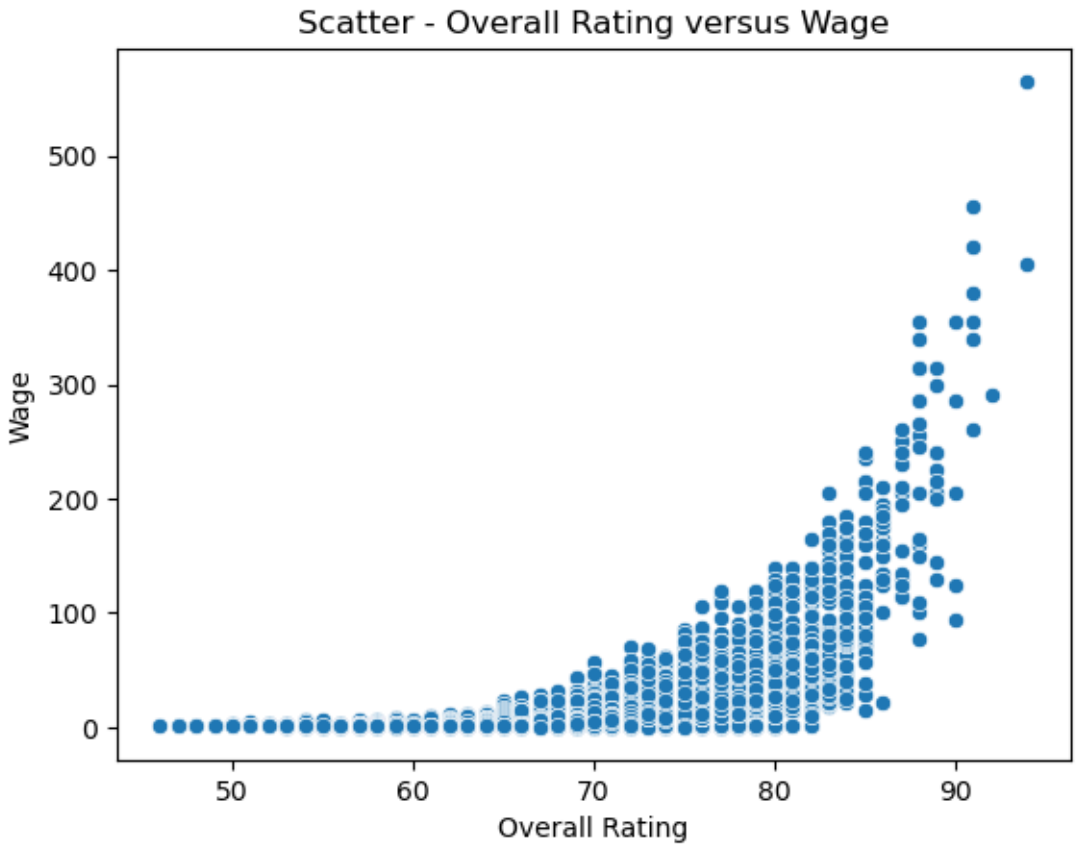
1.1.2 Scatter Plot

[5]:
```
# Scatter Plot of 'Overall' versus 'Wage' using Matplotlib
print(df[['Overall', 'Wage']].head())
plt.scatter(df[['Overall']], df[['Wage']])
plt.title('Scatter - Overall Rating versus Wage')
plt.xlabel('Overall Rating')
plt.ylabel('Wage')
plt.show()
```

```
   Overall   Wage
0       94  565.0
1       94  405.0
2       92  290.0
3       91  260.0
4       91  355.0
```



[6]:
```
# Scatter Plot of 'Overall' versus 'Wage' using Seaborn
sns.scatterplot(data = df, x = 'Overall', y = 'Wage')
plt.title('Scatter - Overall Rating versus Wage')
plt.xlabel('Overall Rating')
plt.ylabel('Wage')
plt.show()
```



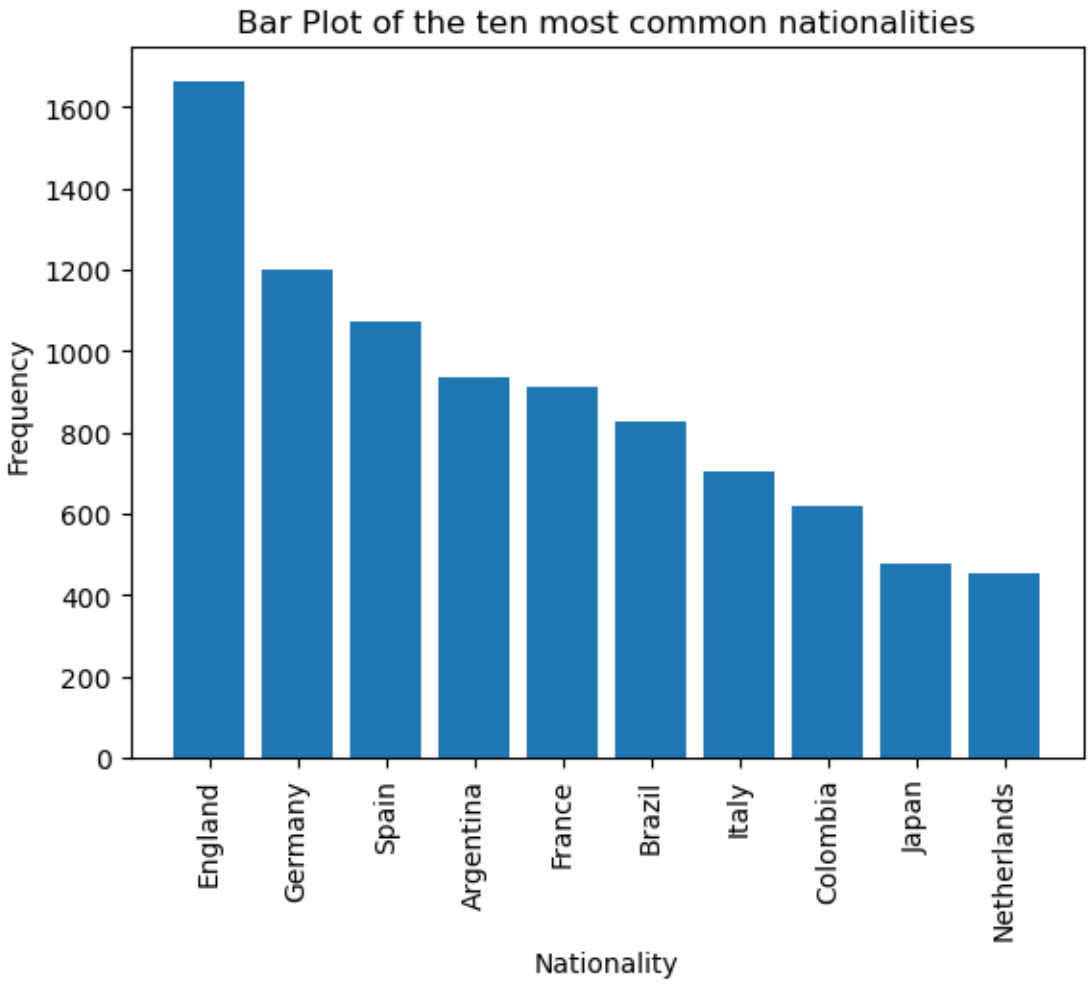
1.1.3 Bar Plot

[7]:
```
# Determining que quantity of each nationality
# Notice that 'Counter' requires a 'Series' instead of a 'Dataframe'
from collections import Counter
print(type(df['Nationality']))
print(dict(Counter(df['Nationality']).most_common(10)))
```

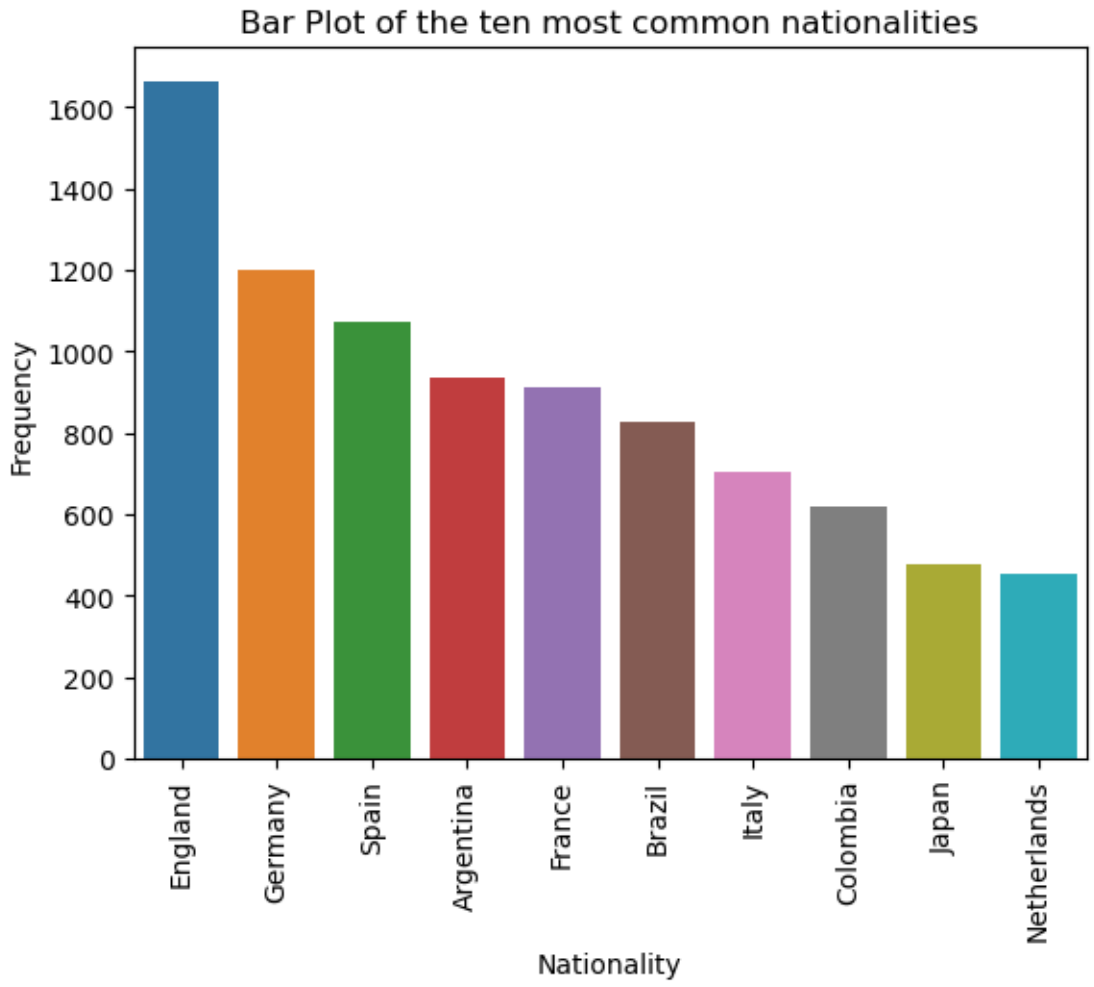
```
<class 'pandas.core.series.Series'>
{'England': 1662, 'Germany': 1198, 'Spain': 1072, 'Argentina': 937, 'France':
914, 'Brazil': 827, 'Italy': 702, 'Colombia': 618, 'Japan': 478, 'Netherlands':
453}
```

[8]:
```
# Bar Plot of the top 10 nationalities using Matplotlib
nationality_top10 = dict(Counter(df['Nationality']).most_common(10))
plt.bar(nationality_top10.keys(), nationality_top10.values())
plt.title('Bar Plot of the ten most common nationalities')
plt.xlabel('Nationality')
plt.xticks(rotation = 90)
plt.ylabel('Frequency')
```

```
plt.show()
```



[9]:
```
# Bar Plot of the top 10 nationalities using Seaborn
nationality_top10 = dict(Counter(df['Nationality']).most_common(10))
df_nationality_top10 = pd.DataFrame([nationality_top10])
df_nationality_top10
sns.barplot(data = df_nationality_top10)
plt.title('Bar Plot of the ten most common nationalities')
plt.xlabel('Nationality')
plt.xticks(rotation = 90)
plt.ylabel('Frequency')
plt.show()
```



1.1.4 Pie Chart

[10]:
```
# Creating a new column 'Main Nationalities' indicating the main nationalities
# In case an unwanted column is added, it can be removed as follows
#df.drop(columns = ['main_nationalities'], inplace = True)
df.loc[df['Nationality'] == 'England', 'Main Nationalities'] = 'England'
df.loc[df['Nationality'] == 'Germany', 'Main Nationalities'] = 'Germany'
df.loc[df['Nationality'] == 'Spain', 'Main Nationalities'] = 'Spain'
df.loc[df['Nationality'] == 'Argentina', 'Main Nationalities'] = 'Argentina'
df.loc[df['Nationality'] == 'France', 'Main Nationalities'] = 'France'
df.loc[df['Nationality'] == 'Brazil', 'Main Nationalities'] = 'Brazil'
df.loc[~df['Nationality'].isin(['England', 'Germany', 'Spain', 'Argentina',␣
 ↪'France', 'Brazil']),  'Main Nationalities'] = 'Other'
df.head()
```

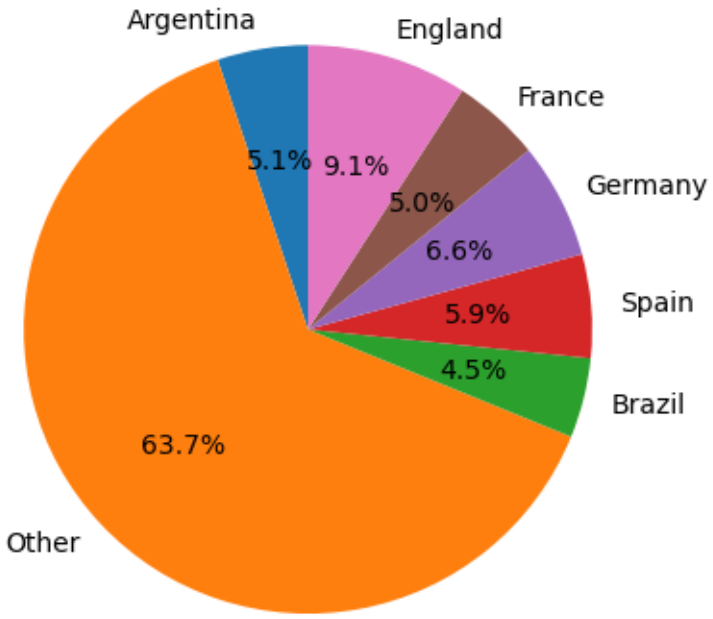
[10]:

	ID	Name	Age	Nationality	Overall	Potential	Club	Value	Wage	Preferred Foot	International Reputation	Skill Moves	Position	Joined	Contract Valid Until	Height	Weight	Release Clause	Main Nationalities
0	158023	L. Messi	31	Argentina	94	94	FC Barcelona	110500.0	565.0	Left	5.0	4.0	RF	2004	2021-01-01	5.583333	159.0	226500.0	Argentina
1	20801	Cristiano Ronaldo	33	Portugal	94	94	Juventus	77000.0	405.0	Right	5.0	5.0	ST	2018	2022-01-01	6.166667	183.0	127100.0	Other
2	190871	Neymar Jr	26	Brazil	92	93	Paris Saint-Germain	118500.0	290.0	Right	5.0	5.0	LW	2017	2022-01-01	5.750000	150.0	228100.0	Brazil
3	193080	De Gea	27	Spain	91	93	Manchester United	72000.0	260.0	Right	4.0	1.0	GK	2011	2020-01-01	6.333333	168.0	138600.0	Spain
4	192985	K. De Bruyne	27	Belgium	91	92	Manchester City	102000.0	355.0	Right	4.0	4.0	RCM	2015	2023-01-01	5.916667	154.0	196400.0	Other

[11]:
```
# Determining the proportions for the Pie Chart
print(Counter(df['Main Nationalities']))
pie_proportions = dict(Counter(df['Main Nationalities']))
print(pie_proportions)
print(pie_proportions.items())
for key, value in pie_proportions.items():
    pie_proportions[key] = (value/len(df))*100
print(pie_proportions)
```

```
Counter({'Other': 11597, 'England': 1662, 'Germany': 1198, 'Spain': 1072,
'Argentina': 937, 'France': 914, 'Brazil': 827})
{'Argentina': 937, 'Other': 11597, 'Brazil': 827, 'Spain': 1072, 'Germany':
1198, 'France': 914, 'England': 1662}
dict_items([('Argentina', 937), ('Other', 11597), ('Brazil', 827), ('Spain',
1072), ('Germany', 1198), ('France', 914), ('England', 1662)])
{'Argentina': 5.146372274399956, 'Other': 63.69528203438238, 'Brazil':
4.542209040478936, 'Spain': 5.887845334212116, 'Germany': 6.579886856703466,
'France': 5.020047234580106, 'England': 9.12835722524304}
```

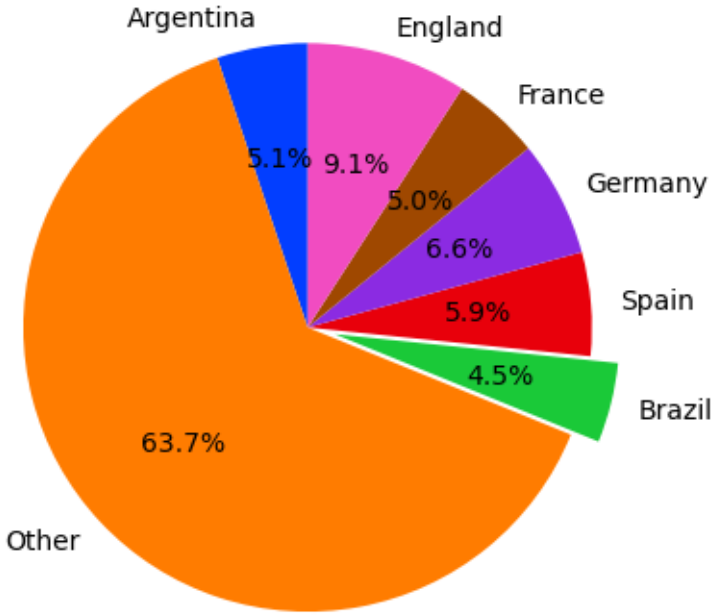
[12]:
```
# Pie Chart of the main nationalities using Matplotlib
fig, ax = plt.subplots()
ax.axis('equal')
ax.pie(pie_proportions.values(), labels = pie_proportions.keys(), autopct='%1.
 ↪1f%%', startangle = 90)
plt.show()
```



[13]:
```
# Pie Chart of the main nationalities using Matplotlib and Seaborn
df_pie_proportions = pd.DataFrame([pie_proportions])
print(df_pie_proportions)
df_pie_explode = pd.DataFrame([{'Argentina': 0, 'Other': 0, 'Brazil': 0.1,␣
 ↪'Spain': 0, 'Germany': 0, 'France': 0, 'England': 0}])
print(df_pie_explode)
df_pie_proportions = pd.concat([df_pie_proportions, df_pie_explode],␣
 ↪ignore_index = True)
print(df_pie_proportions)
palette = sns.color_palette('bright')
plt.pie(df_pie_proportions.loc[0], labels = list(df_pie_proportions), colors =␣
 ↪palette, explode = df_pie_proportions.loc[1], autopct='%1.1f%%', startangle␣
 ↪= 90)
plt.show()
```

```
   Argentina      Other     Brazil      Spain    Germany     France    England
```

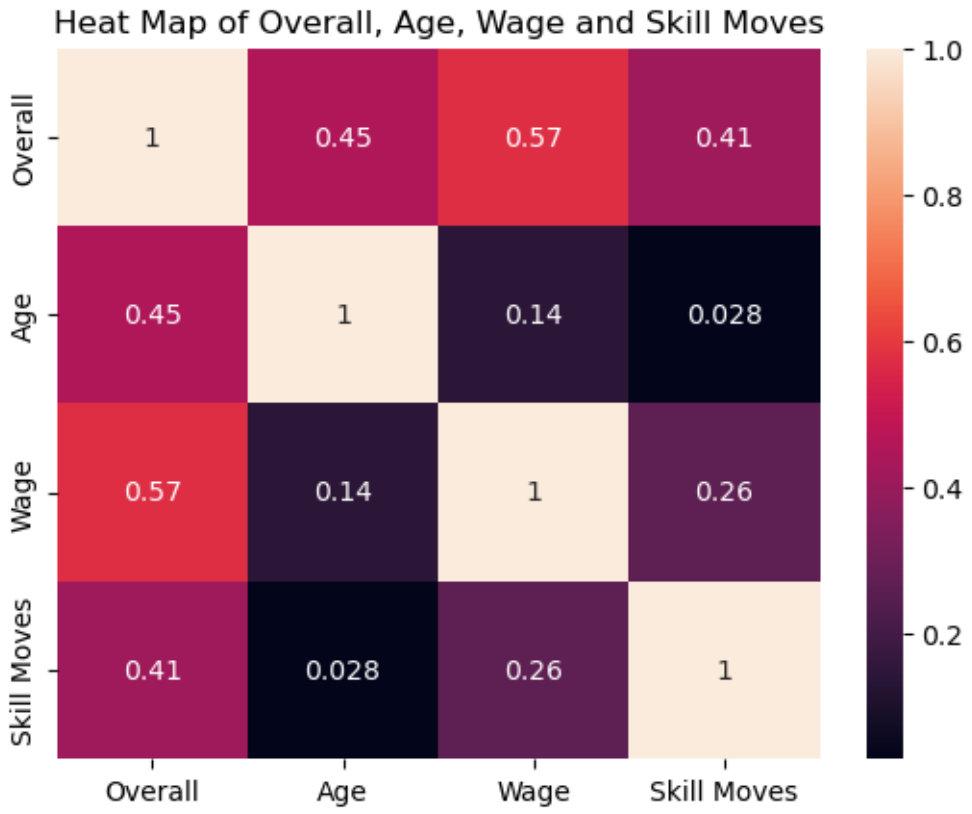
```
0   5.146372  63.695282  4.542209  5.887845  6.579887  5.020047  9.128357
   Argentina  Other  Brazil  Spain  Germany  France  England
0          0      0     0.1      0        0       0        0
   Argentina      Other    Brazil     Spain   Germany    France   England
0   5.146372  63.695282  4.542209  5.887845  6.579887  5.020047  9.128357
1   0.000000   0.000000  0.100000  0.000000  0.000000  0.000000  0.000000
```



1.1.5 Heat Map

[14]:
```
# Heat Map of 'Overall', 'Age', 'Wage', 'Skill Moves' using Seaborn
correlation = df[['Overall', 'Age', 'Wage', 'Skill Moves']].corr()
print(type(correlation))
print(correlation)
sns.heatmap(correlation, annot = True)
plt.title('Heat Map of Overall, Age, Wage and Skill Moves')
plt.show()
```

```
<class 'pandas.core.frame.DataFrame'>
              Overall       Age      Wage  Skill Moves
Overall      1.000000  0.452350  0.571926     0.414463
Age          0.452350  1.000000  0.141145     0.027649
Wage         0.571926  0.141145  1.000000     0.263205
Skill Moves  0.414463  0.027649  0.263205     1.000000
```



2 Visualisation Practice Exercise 2

2.1 Visualize Distributions With Seaborn

This practice exercise was guided by this tutorial.

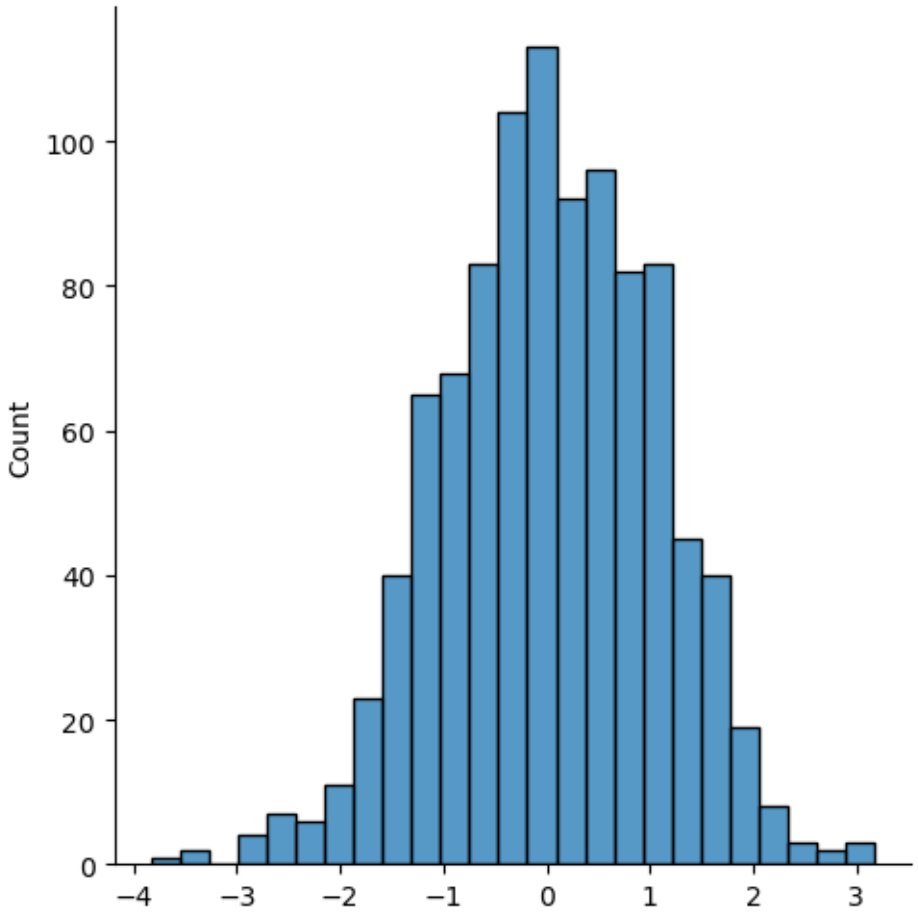
[15]:
```
# Importing the required libraries
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
import seaborn as sns
```

2.1.1 Normal distribution

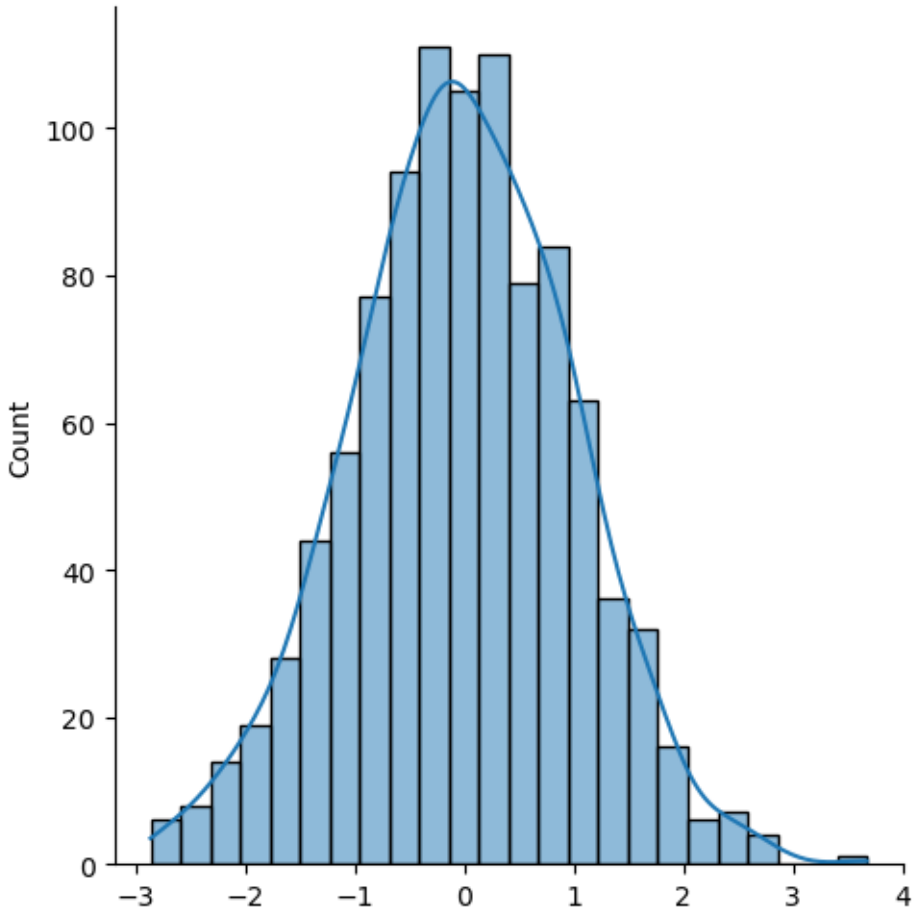
[16]:
```
# Trying out Numpy's 'random' function
normal_distribution = np.random.normal(size = (2, 3))
print(normal_distribution)
```

```
[[ 1.3460287  -0.5492698   0.7147615 ]
 [ 0.11518817  0.40616814 -0.87478943]]
```

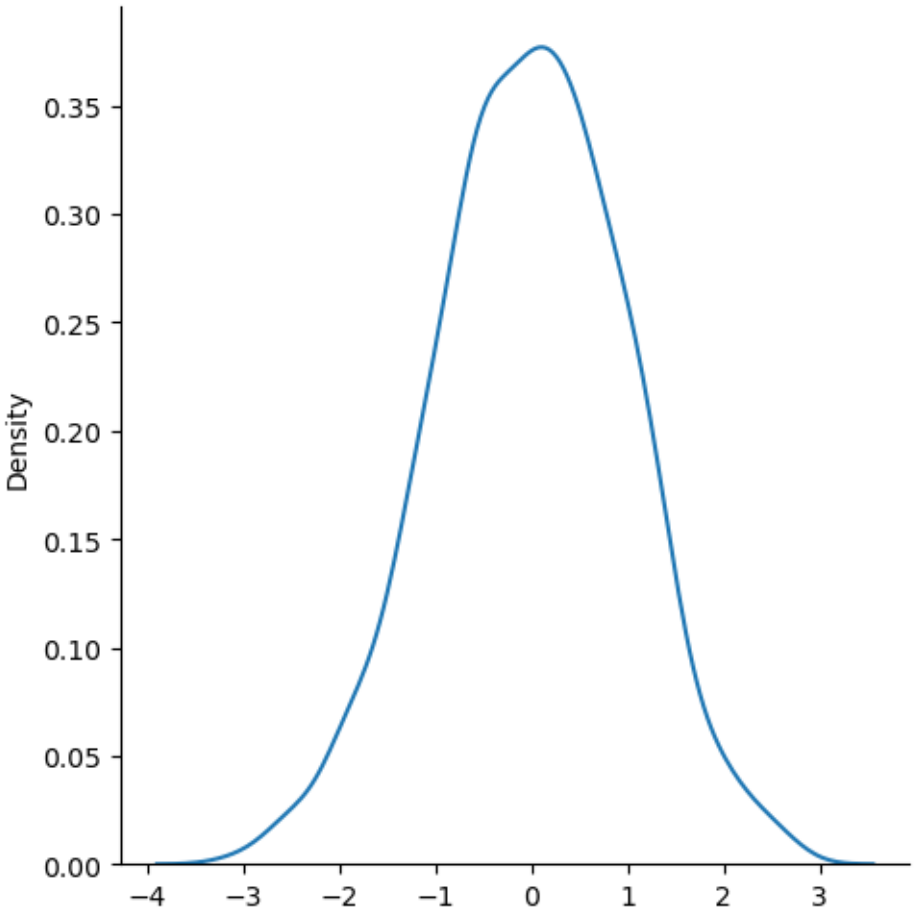
[17]:
```
# 'displot' defaults to 'histplot' when the parameter 'kind' is not specified
sns.displot(np.random.normal(size=1000))
plt.show()
```



[18]:
```
sns.displot(np.random.normal(size=1000), kde = True)
plt.show()
```



[19]:
```
sns.displot(np.random.normal(size=1000), kind = 'kde')
plt.show()
```


Density
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
−4
−3
−2
−1
0
1
2
3

[ ]: